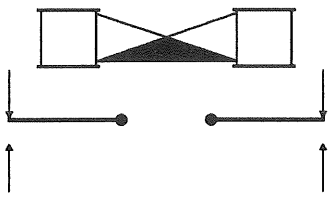
INTRODUCTION TO SIGNALLING

7

RELAYS AND RELAY LOGIC



A.J.R.Rowbotham

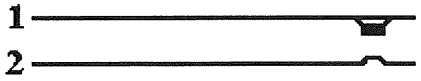
INTRODUCTION

For many decades before the advent of SSI, the interlocking function was carried out by relay circuitry. An interlocking consisted of many hundreds or even thousands of relays appropriately interconnected to put together the safety logic conditions required. Although SSI is installed for all new works in the UK, the rudiments and background to the signalling philosophy cannot be really learnt from this technology, as there is 'nothing to look at'. Relay circuits can be studied, however, without difficulty, and with a small number of exceptions, the SSI working replicates relay logic.

THE RELAY

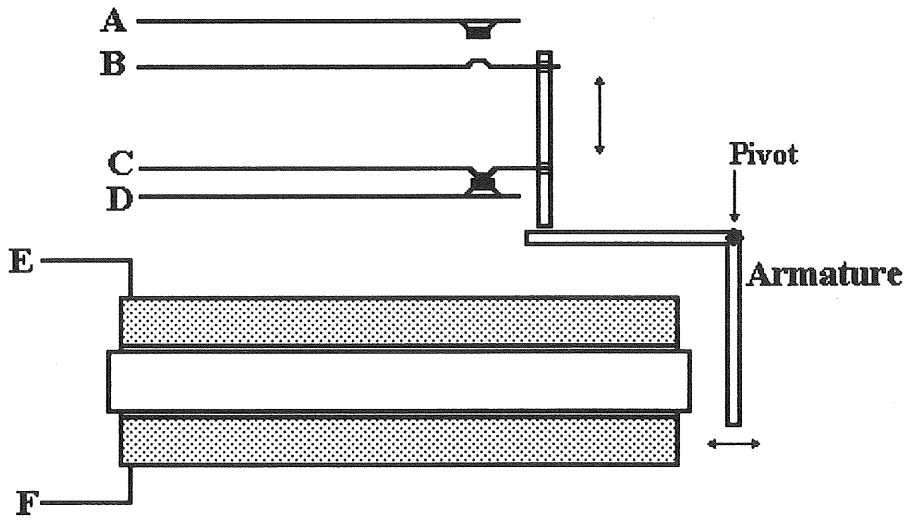
A relay consists of a number of switches, or *contacts*, and a coil. The contacts may all be operated simultaneously by applying a voltage to the coil.

Each contact consists of two flexible contact arms, one of which is fitted with a pad of graphite impregnated with silver.



The pair of arms act as a switch, if the two are not in contact, as in the diagram above, the contact is open and no current can flow through the contact. If the two are pushed together so they touch (at the right hand end in the diagram) the contact will be closed and current can flow.

The movement of the arms is provided by the action of an electromagnet, formed by a metal core within the coil of the relay. The diagram below shows the principle of this action. It is assumed in this diagram that no current is flowing through the coil E/F. At the top are two contacts - A/B is shown open, whilst C/D is shown closed.



If a voltage is applied to the coil then the resultant magnetic force attracts the pivoted armature, which in turn pushes up two of the contact arms. This results in contact A/B being closed and contact C/D being opened. The state of the contacts can thus be controlled by the state of the coil.

Nomenclature

A relay is said to be *energised* if the current is flowing through the coil, and *de-energised* if it is not. These are the formal terms, but other terms are commonly used. A relay is said to have picked up when current is applied to the coil, and to have dropped away when the current is removed. The terms *picked* (for energised) and *dropped* (for de-energised) are often used, as are the even simpler plain *up* (for energised) and *down* (for de-energised).

It will be seen from the diagram above that there are two types of contacts involved. In other professions it is common to use the terms 'normally open' and 'normally closed' to describe contacts A/B and C/D respectively, assuming that the relay is normally de-energised and picks to fulfil its function. However, to ensure the required safety, it is common in signalling for some relays to be normally energised, and to drop to fulfil their function. The term 'normally' thus becomes meaningless and different terms have to be used. Like a lot of things in signalling, the terms are now anachronistic but when they were introduced were meaningful. The type of contact closed when the relay is de-energised (C/D in the diagram) is known as a *back contact* as in early relay design they were at the back of the relay. Conversely, the type of contact that is closed when the relay is energised (A/B in the diagram) is known as a *front contact.*

Types of Relay

The current design of relay is known as the *BR Miniature* relay or the *BR 930 series* relay. BR 930 is the specification for the basic relay, and there are over twenty other specifications for the various types in the family.

The different types of relay include:

- *Neutral* - the basic relay with no special features.
- *Immune* - designed to operate with d.c. voltages (normally 50 V), but not on a.c. voltages.
- *Biased* - magnetically designed so that the relay will only operate when the current is passed through the coil in one direction. Current flowing in the opposite direction will not operate the relay.
- *Contactor* - fitted with heavy duty contacts, usually for point machine operation. The contacts have a magnetic blow-out feature to stop arcing when they open whilst carrying a heavy current.
- *Latched* - fitted with two coils, one of which energises the relay, but once 'energised' the current may be removed from the coil without affecting the contacts, which are magnetically latched. The relay may be unlatched by either energising the second coil, or by energising the first coil in the opposite direction.
- *Double wound* - fitted with two independent coils, either of which will operate the relay.
- *Slow pick-up* - magnetically designed so that the contacts do not start to open/close directly the relay is energised.
- *Slow release (*or *slow drop away)* - magnetically designed so that the contacts do not start to open/close directly the relay is de-energised.
- *Track Relay* - specially designed with a low resistance coil to work as part of a track circuit.
- *Twin Relay* - two independent relays housed within the standard relay case.

The maximum number of contacts possible on a standard BR Miniature relay is 16, arranged in four columns of four. One family of relays is based on a combination of eight fronts and eight backs ('8F/8B'). In general many more front contacts are needed than backs, so another family is based on a combination of twelve front and four back contacts ('12F/4B'). Twin relays have a maximum of eight contacts on each relay, basically in a 4F/4B or 6F/2B configuration. The different types of relay described above do not necessarily have a complete set of 16 contacts, but would be based on the above configurations, with say one row

or one column of contacts missing. The connections to the contacts and coil are numbered on a co-ordinate system. The columns of contacts are designated A, B, C and D, and within each column the connections are numbered. The two connections to a contact can thus be numbered as A1/A2, D7/D8 etc. The coil connections are designated by the letter R, so a typical coil connection would be R1/R2.

All relays plug in to a standard plugboard mounted on suitable racking. The wiring is held in the plugboard, so replacing a relay does not disturb the wiring. Each wire is terminated with a crimped connector, the *crimp*, which, when inserted in the plugboard, locks in position. A special tool is needed to release the crimp from the plugboard when a wire has to be removed.

It is essential that the type of relay specified in the diagrams is plugged into the relevant plugboard. If a 8F/4B relay was plugged in instead of a 12F/8B relay, then four of what should be front contacts would actually be back contacts, with extremely dire consequences, such as the rear of a train not being protected by a red signal. Every relay is fitted with five pins, arranged in a unique pattern for each relay type. Each plugboard has to have the equivalent positions drilled out relative to this relay *pincode* before the relay can be plugged in.

RELAY CIRCUIT DIAGRAMS

In signalling practice, the same circuit diagram is used for a number of activities. There is no requirement to draw different forms of the same circuit for different activities. These activities include:

- design.
- checking.
- installation.
- testing.
- commissioning.
- maintenance.
- decommissioning.

The same diagram must include all the information for these activities, so all diagrams are actually true wiring diagrams rather than schematics.

Symbols

Three symbols are required to draw relay components - the coil, the front contacts and the back contacts. The symbols used are specified in BS 376, and are unique to the signalling profession. They are quick to draw and produce tidy, easy to read, circuits - essential when there is a failure at Waterloo in the rush hour.

The symbols have been designed so that they may all be added to a straight line drawn across the page. There is no need to allow for gaps in the line for symbols to be added later.

The symbol for a relay coil (and in fact for other items as well), is a rectangle hung underneath the line.

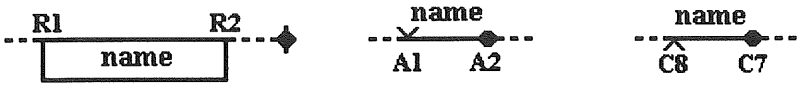


The symbol for a contact is a 'V' and a 'dot' added to the line.

The 'V' is upright for a front contact, and upside-down for a back contact.



The name of the function that a relay is performing is inserted inside (or sometimes above) the rectangle representing the coil, and above the contact symbol. The contact and coil numbers are also shown:



A series of letter codes are used, that when applied in combination, describe uniquely the function of the relay. All contacts of a given relay will have the name of that relay. On a signalling scheme there must not be two relays with the same 'name'.

Two other symbols are necessary to complete simple circuits - the positive and negative sides of the power supply. Normally circuits are drawn with the positive side of the supply on the left, and the negative side on the right, so the current flows across the page from left to right. The positive side is fused, the symbol for which is:



whilst the negative supply is just a terminal on a bus-bar:



Simple Logical Circuits

The circuits to be described are not necessarily practical circuits, but are chosen to show the logic utilised.

Example 1

Suppose it is required that a relay R is controlled by relay A, to the requirement:

Relay R to be energised only when relay A is energised

This statement could equally have been written as

Relay R to be de-energised only when relay A de-energised

Both statements represent the same logic and result in the same circuit. A logic expression defining a circuit can either define the conditions for energising the relay, or for de-energising the relay.

To start, draw a line across the page with the power supply symbols at the ends:



The requirement is to control relay R by contacts of relay A, so the coil of relay R can immediately be added towards the right of the circuit.



Between the fuse and the coil a contact of relay A is required, so that the logic detailed above is achieved. A decision needs to be made - is it a front contact of relay A that is required, or a back contact?

For relay R to be energised, the current must be flowing, so the chosen contact of relay A must be closed to complete the circuit. The requirement is for relay A to be energised to energise relay R, so the chosen contact of relay A must be closed when the relay is energised, i.e. a front contact.



This completes the circuit - a front contact of relay A controls relay R. In this situation, relay R is said to be a *repeater* of relay A, as it behaves in the same way.

Example 2

Suppose it is required that a relay R is controlled by relay A, to the requirement:

Relay R to be energised only when relay A is de-energised

This statement could equally have been written as

Relay R to be de-energised only when relay A energised

Again. both statements result in the same circuit.

Using the same arguments, the current to energise relay R must flow through a closed contact of relay A, this time only while relay A is de-energised - so a back contact.



[In any case there are only two types of contact to chose from, so if it is not a front contact, it must be a back contact!]

Example 3

In this exercise, three relays are involved. The logic required is:

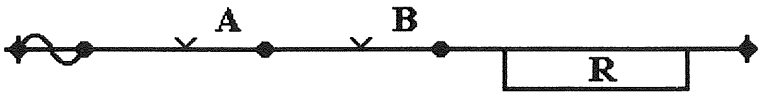
Relay R to be energised only when relay A AND relay B are both energised

To write this the other way, starting with 'Relay R to be de-energised...' is not quite so easy. The correct logical expression is actually:

Relay R to be de-energised only when relay A OR relay B is de-energised

Note that the AND in the 'energised' requirement changes to OR in the 'de-energised' expression. In signalling logic it is understood that the 'OR' condition above does include both relays de-energised as well, i.e. it is an 'inclusive OR'.

In this example there will be a contact of relay A and a contact of relay B in the circuit for relay R. In order for the current to flow, all contacts of relay A and relay B in the circuit must be closed, and only when relay A and relay B are both energised, so a front contact of relay A *in series* with a front contact of relay B is required:



Note that both relays A and B have to be energised for the current to flow, but either relay de-energised (or both) will break the circuit. Swapping the position of the two contacts round would not alter the behaviour of the circuit - the position of contacts in a series circuit is not important in theory.

When drawing circuits, the distance of the gap between the contacts should be greater than the length of the contact so it is clear which 'dot' belongs to which 'V'.

Example 4

Again with three relays. The logic required is:

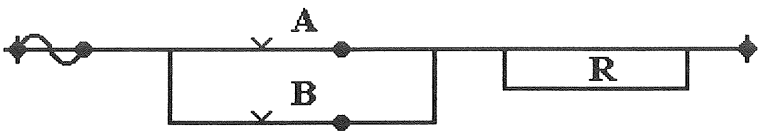
Relay R to be energised only when relay A OR relay B is energised

Or, writing the same logic as a 'de-energise' expression:

Relay R to be de-energised only when relay A AND relay B are both de-energised

Note the change from OR to AND to write the 'de-energised' expression.

For this circuit the OR means that alternative paths are required to energise relay R, so the contacts of relays A and B need to be *in parallel*, and schematically the circuit would be:

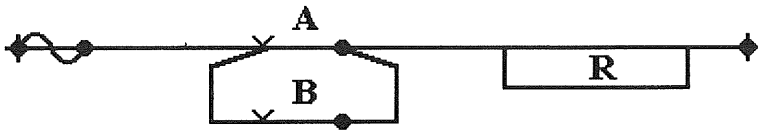


Note that either of relays A and B (or both) have to be energised for the current to flow, but only both relays de-energised will break the circuit. Swapping the position of the two contacts round would not alter the behaviour of the circuit - the position of contacts in a parallel circuit is not important in theory.

As drawn above the circuit is shown schematically, it is not a wiring circuit - how does the installer join the wires together to make the parallel path? The design of the relay wiring crimps allows for two wires to be inserted. Thus one way of joining wires is by a relay crimp. Also, two wires (and not more) may be terminated on fuses and power terminals. Thus in the schematic diagram above, the wire on the left hand side of the relay B contact could be connected as a second wire to the fuse, or a second wire to the left hand side of the contact of relay A. Similarly, the wire on the right hand side of the relay B contact could be connected as a second wire to the left hand end of the coil of relay R, or a second wire to the right hand side of the contact of relay A.

Conventionally, and unless there was a good reason to do otherwise, second wires would be

added to both sides of the contact of relay A. On diagrams, the second wire is shown at an angle for a short distance:



Summary of Logic

All logic is built up from ANDs and ORs resulting in series or parallel circuits. The relationship can be summarised in a table:

	AND CONDITION	*OR CONDITION*
To energise a relay	contacts in series	contacts in parallel
To de-energise a relay	contacts in parallel	contacts in series

Further Simple Logical Circuits

Example 5

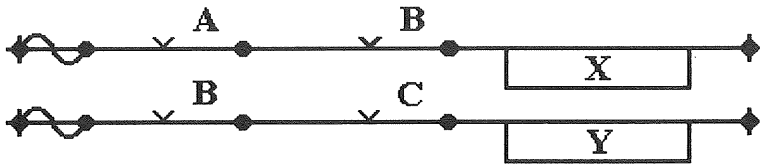
Consider two requirements:

Relay X to be energised only when relay A AND relay B are both energised

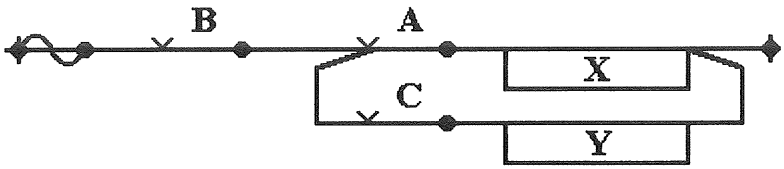
and

Relay Y to be energised only when relay B AND relay C are both energised

At first sight the two circuits would be:



However, by altering the order of series contacts in one of the circuits, the two circuits may be economically combined:



Example 6

This exercise introduces a new and important concept. Consider the requirement:

Relay R to be energised when a button is pushed, and to remain energised when button is released

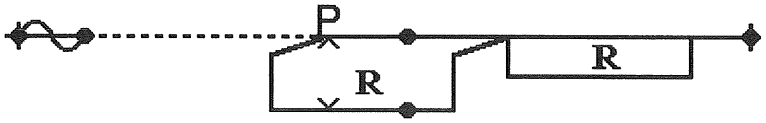
The symbol for a normal 'push to make' button is



The circuit can be started thus:



so relay R will be energised as long as the button is being pushed, but will de-energise directly the button is released. An alternative path needs to be provided to the coil of relay R, but only once the relay is energised. This path can be provided by a front contact of relay R itself, in parallel with the push button contact:



In the quiescent state relay R will be de-energised, so the front contact will be open. Consequently, the only path to the coil is through the push button. Directly relay R is energised, the front contact will close to provide the alternative path, keeping the relay energised when the push button contact is no longer closed. Of course, as shown, the relay, once energised, will stay energised for ever, so it is assumed there are other contacts, not relevant to the argument, in the 'dotted section' of the circuit, which will be used to de-energise relay R. This type of requirement, where the conditions necessary to energise a relay can alter, once the relay has energised, is very common. It can be looked on as registering that a certain condition has occurred - in this case that a button has been pushed.

This sort of circuit is known as a *'stick'* circuit - the relay 'picks and sticks'. The front contact of relay R used is consequently known as the *'stick contact'*. It is normal practice to show the relay side of the stick contact connected as a second wire to the relay coil, rather than, in this example, a second wire on the push button contact. It is in any case a short loop of wire on the plugboard of relay R, rather than possibly a long wire between the relay and the push button.

FUNCTION NAMES

A collection of letter codes are used to indicate the function of the items represented in circuit diagrams. The form of this designation is usually the identity of the item (i.e. usually the point or signal number, or the track circuit alpha reference), followed by a number of letters defining the function related to the item. The last letter in the name indicates what the item is, using the following letters:

R Relay

E Lamp

I Inductor

M Motor

Note that 'R' has other meanings when it is not this last letter, whilst 'E' has the same meaning wherever it is used.

The same symbol, the rectangle, represents any of the following: - a relay coil, a lamp, an inductor (AWS magnet) or a motor. Only this right-hand-most letter details exactly which it is.

The letters preceding this last letter act like adjectives, giving more detail. Letters used in these positions include:

C Proving (or checking)

G Signal

K Indicating or detection

P Repeater

T Track circuit

W Point

For instance:

CR Proving relay

GE Signal lamp

KE Indicating lamp

TR Track Relay

WM Point motor

Further letters add more detail if required:

D Green

H Yellow

N Normal

R Red or reverse

U Route

Typical and common names encountered include:

AJR TR	Track relay for AJR track circuit
AJR TKE	Track indication lamp for AJR track circuit
23 ECR	Lamp proving relay for 23 signal
23 RGE	Red lamp of 23 signal
23 HGE	(first) yellow lamp of 23 signal
23 HHGE	Second yellow lamp of 23 signal
23 UGE	Lamps of the JDI (Route arm) attached to 23 signal
23 I	AWS inductor for 23 signal
345 NWKR	Normal detection relay for 345 points
345 RWKR	Reverse detection relay for 345 points
345 RWKE	Reverse detection indicating lamp for 345 points

Although the letter 'R' represents 'relay', 'red' or 'reverse', there is usually no confusion as in the examples above, the letter 'R' is associated with either a 'G' for signal (inferring 'red') or a 'W' for points (inferring 'reverse'), or is the last letter (for 'relay'). In any case, these very common names become familiar with experience.

Example 1 above developed the circuit for a repeater relay. The letter 'P' is used in the name to indicate such a function:

AJR TR	Track relay for AJR track circuit
AJR TPR	Repeater relay for the track relay of AJR track circuit, usually termed 'track repeater relay'

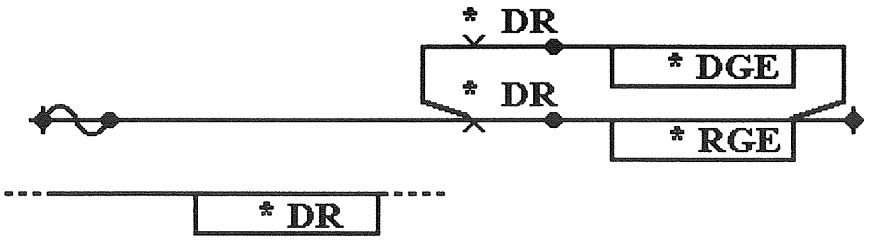
The examples above are a small selection of the letters used. For a more detailed list refer to BS 376.

PRACTICAL CIRCUITS

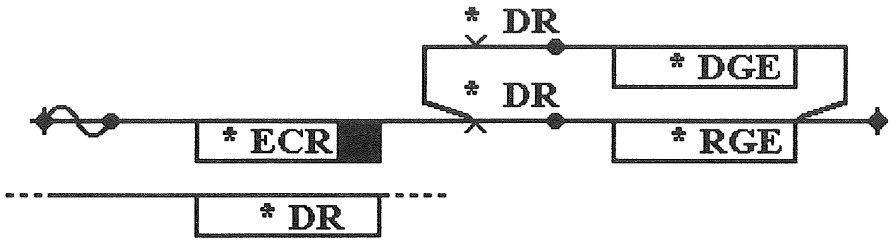
2-aspect Signal

Consider a simple 2-aspect signal showing either red or green. It may be considered to be a 2-state device, i.e. red or green. A relay is also a 2-state device - energised or de-energised. Consequently it can be deduced that one relay is sufficient to control one 2-aspect signal. However, a decision has to be made as to 'which way up' to use the relay - i.e. energised to light the green, or energised to light the red? Any failures will result in the controlling relay being de-energised, so to be safe, the relay must be de-energised when lighting the red lamp, and consequently energised to light the green.

As a result, the relay is said to be controlling the green aspect, and is named DR as a result (note it is not DGR for some reason). It is of no concern for this exercise what controls there are in the DR circuit. The circuit can be started (where the * represents the signal number):



(**Please note** that these circuits are being discussed to show the logic and the use of symbols, not the true and full detail of the circuit - for instance there are transformers between the lamps and the controlling contacts in practice.) It is an interlocking requirement to continuously prove that the signal is lit, and that a lamp has not failed. Incandescent filament lamps are used, which only fail by burning out to an open circuit, never to a short circuit. Hence if there is a current flowing into the signal, it can safely be assumed that a lamp is lit. To prove this current flow, it is necessary to put a relay coil in the circuit. If a current is flowing, then this relay will remain energised. This relay is known as the *lamp proving relay,* ECR. This relay is virtually permanently energised (assuming the lamps do not fail), however there is an instant when the aspect is changing when no current is flowing (in the period between one contact breaking and the other making). In order to stop the ECR momentarily de-energising every time a signal changes aspect, a special type of relay, a slow drop away relay is used. This relay has a special symbol for its coil to represent this property:



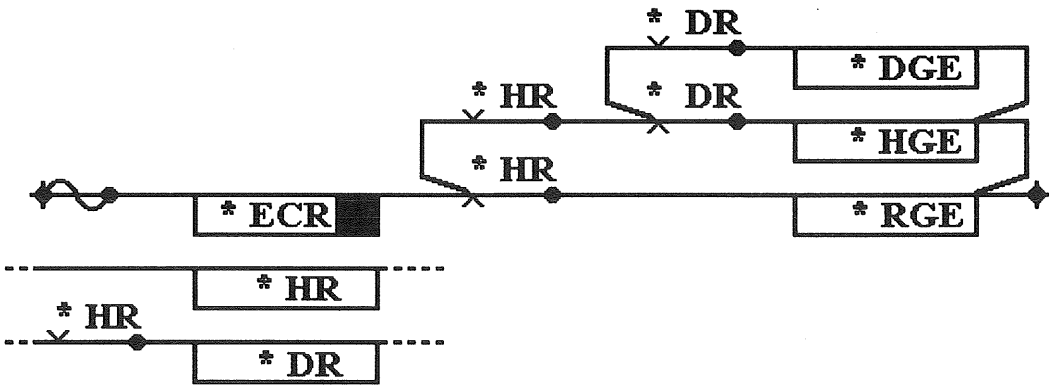
It is specially designed so that when current stops flowing through the coil, there is a delay until the contacts start moving. This delay is such that the current will start again through the

other lamp before any movement takes place. In this way the relay stays energised whilst the aspects are changing, 'filtering out' the changeover gap in current flow.

Another special design feature of lamp proving relays is that they must be of low resistance as they are not the main electrical load. If a high proportion of the circuit voltage was dropped across the relay, then there would not be sufficient voltage to light the lamps.

3-aspect Signal

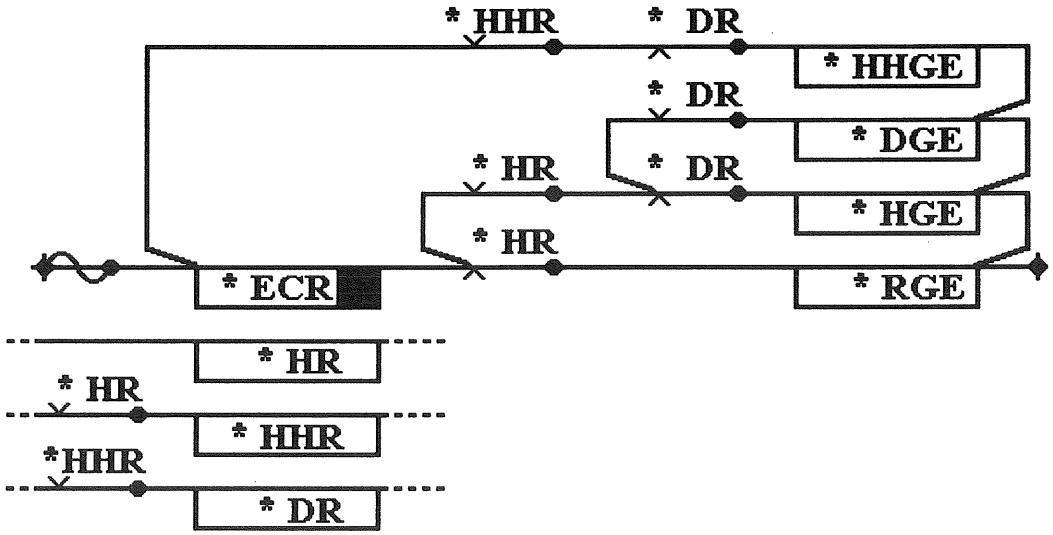
A 3-aspect signal needs two controlling relays. The first controls the yellow aspect, and is consequently called the HR and the second controls the green aspect, so again is called the DR. When the signal is required to be showing red, both relays are de-energised. If the HR is energised the signal shows yellow, and then if the DR is energised as well the signal shows green. To ensure the correct sequence, the DR cannot become energised unless the HR is already energised.



The HR decides whether the signal shows red or one of the proceed aspects, and the DR decides which that aspect is. The 'shape' of the circuit is like a staircase, where the red aspect is at the bottom level and the green aspect at the top level.

4-aspect Signal

The 4-aspect signal requires 3 relays to control it, in order of operation, the HR, the HHR and DR. Again the HR is energised first to show a proceed aspect, the HHR decides whether a yellow or double is shown, and the DR decides whether a double yellow or a green is shown. The relays are forced to energise, as before, in the correct sequence The 'ladder' effect no longer applies as there are two lamps lit together for one of the aspects. However, the 4-aspect circuit is obtained solely by adding a new piece of circuitry to the 3-aspect circuit, rather than changing it. If the signal is showing yellow and the HHR is subsequently energised, the first yellow is not affected whilst the second yellow is illuminated. If the DR is then energised the feeds to both yellow lamps are broken by back contacts of the DR, and the signal changes to the green aspect.



Note that it is not considered necessary to lamp prove the second yellow lamp. If it fails a more restrictive aspect is displayed, and in any case, the lamp proving relay is carefully designed to monitor the current expected with one lamp.